



IRAN OPEN ROBOCUP JUNIOR SOCCER SIMULATION 2025



TEAM DESCRIPTION PAPER

Danesh



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Abstract

This Team Description Paper (TDP) provides a comprehensive overview of the software of simulated robots participating in the RoboCup Junior Soccer Simulation league. The software, As the primary unit of controlling and making decisions, is implemented to handle the dynamic nature of the on-going simulated match and to act appropriately considering the status of the robot and the field. The actions of the robots can be summarized in offensive actions, to lead the ball toward the opposite goal, defensive actions, to obstruct the opponent's progress and to reposition.

- **Software architecture**

- a. **Interacting with the simulation environment**

The simulation is based on the environment, created in Webots applications. The robots created in this application are considered objects that come with multiple attributes. These attributes are the main way of interaction with the simulation. Considering this, it is important to understand how these attributes work and how we can make use of them. These attributes consist of multiple virtual sensors and actuators.

- 1. **Compass sensor**

This virtual sensor gives the current angle of the robot. Using this data, we can track the current angle and control rotations of the robot. We have access to this data using `get_compass_heading()` method.

2. GPS sensor

This virtual sensor allows us to keep the location of the robot, presented in two float numbers as (x , y). By using these numbers, it is possible for the robot to move to different desired locations. We have access to this data using `get_gps_coordinates()` method.

3. Sonar sensors

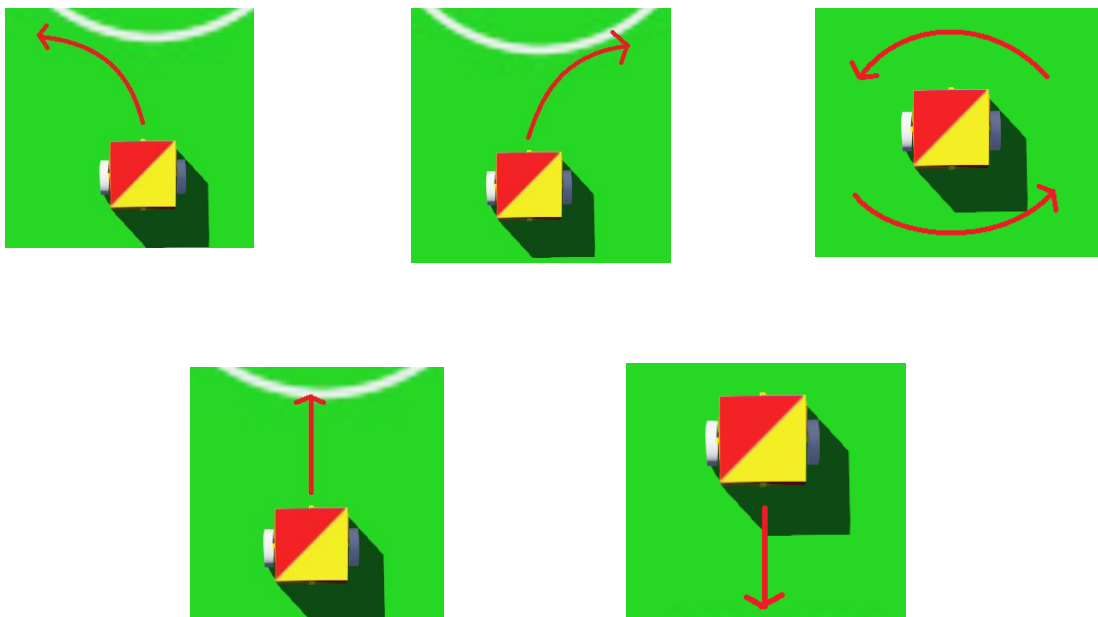
4 virtual distance sensors are placed on 4 sides of the robot. These sensors determine the distance of any obstacle in front of each side of the robot. They will help us to keep a certain distance from the walls or to detect if the robot has collided with any obstacles. By using `get_sonar_values()` method, we can get the collection of 4 sensors.

4. Ball direction data

This data will give us information about the position of the ball relative to the robot. This data has 3 different values: -1 (right), 1 (left) and 0 (front). By using this data, the robot can rotate in the desired direction until the ball is right in front of it.

5. Actuators

2 motors are placed in the robot. We can set speed of each motor individually using `left_motor.setVelocity()` or `right_motor.setVelocity()`. Considering this, various movements (moving forward, backward, rotating) can be implemented.



b. General performance of the robots

After receiving and processing sensors data, each robot should execute an appropriate action with respect to status. Most of the possible actions for the robots can be divided into 2 groups.

1. Offensive actions

Since the main purpose of the robots is to earn points by taking the ball to the opponent's goal, it is important to lead the ball successfully. As the offensive actions, the robots try to push the ball with an appropriate angle, toward the goal. This action requires robots to constantly process compass and ball direction sensors data. After determining the appropriate angle, the robots can set both left and right motors individually and make progress.



Figure 1 Leading the ball

2. Defensive actions

As the opponents lead the ball toward our goal, the robots should make an action and prevent the opponent's progress, as the ball approaches. The robots must process the ball's direction and move toward the ball.

By receiving new data about the direction of the ball, robots can act offensively or defensively. In contrast to those situations, the ball getting out of range of the robots' sensors can leave the robots unaware of the events; so, it is an appropriate action to change the position of the robot. The robots can decide to move to better positions, if they have lost track of the ball.



Figure 2 Obstructing opponent's progress

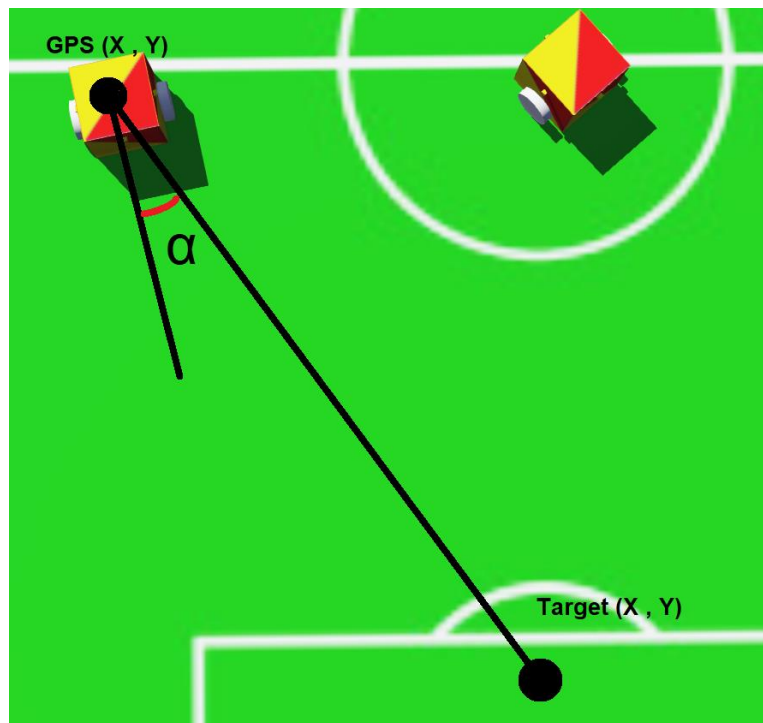
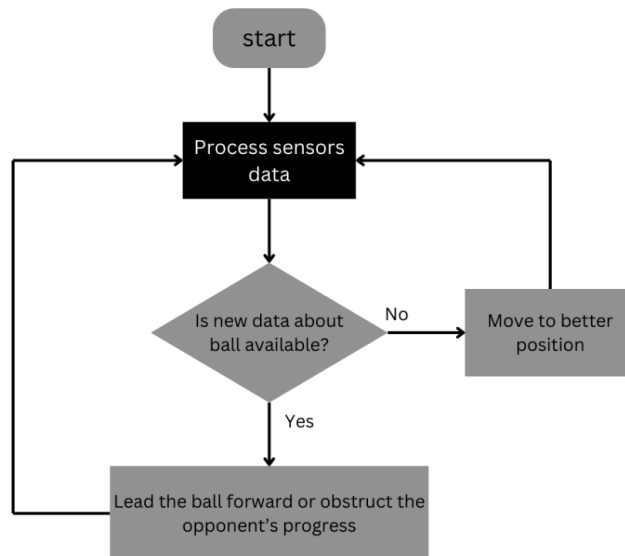


Figure 3 Repositioning

The following diagram shows a general algorithm of the robots.



- **Conclusion**

In conclusion, our software by considering a wide range of states in the on-going match and performing various actions under general terms of offensive and defensive actions, ensures the robots to be prepared for many possible scenarios and to achieve the best performance.

- **Appendix**

<https://robocup-junior.github.io/rcj-soccersim/>